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An Empirical Evaluation of the System Usability Scale

Aaron Bangor (AT&T Labs), Philip T. Kortum (Rice University), and James T. Miller (AT&T Labs)

Published in International Journal of Human-Computer Interaction, 24(6), 574–594 (2008)

DOI: 10.1080/10447310802205776

ABSTRACT

This article presents nearly 10 years' worth of System Usability Scale (SUS) data collected on numerous products in all phases of the development lifecycle. The SUS, developed by Brooke (1996), reflected a strong need in the usability community for a tool that could quickly and easily collect a user's subjective rating of a product's usability. The data in this study indicate that the SUS fulfills that need. Results from the analysis of this large number of SUS scores show that the SUS is a highly robust and versatile tool for usability professionals. The article presents these results and discusses their implications, describes nontraditional uses of the SUS, explains a proposed modification to the SUS to provide an adjective rating that correlates with a given score, and provides details of what constitutes an acceptable SUS score.

AGGREGATE DATA & SAMPLE (p. 576-577)

Since we began collecting data with the SUS in 1996, 2,324 surveys have been completed over the course of 206 studies.

- Individual Surveys (N = 2,324): Mean = 70.14, Median = 75.00, SD = 21.71, Range = 0.00 - 100.0.
- Multisurvey Studies (N = 206): Mean = 69.69, Median = 70.91, SD = 11.87, Range = 30.00 - 93.93.

Table 6: Scores by Interface Type (p. 583):

- Web: Count = 1,180, Mean = 68.05, SD = 21.56.
- Cell Phones: Count = 189, Mean = 66.55, SD = 19.84.
- Customer Premise Equipment (CPE): Count = 219, Mean = 71.60, SD = 21.60.
- Graphical User Interface (GUI): Count = 208, Mean = 75.24, SD = 20.77.
- Interactive Voice Response (IVR): Count = 401, Mean = 73.84, SD = 22.15.
- Web/IVR Combined: Count = 50, Mean = 59.45, SD = 19.19.

ADJECTIVE RATING SCALE PILOT (p. 586-587)

Table 7: Summary Statistics for Adjective Ratings (N = 212):

- Best Imaginable: Count = 1, Mean = 100.0
- Excellent: Count = 69, Mean = 85.58, SD = 9.473
- Good: Count = 90, Mean = 72.75, SD = 10.56
- OK: Count = 36, Mean = 52.01, SD = 12.13
- Poor: Count = 15, Mean = 39.17, SD = 12.38
- Awful: Count = 0, NA
- Worst Imaginable: Count = 1, Mean = 25.0

WHAT IS AN ACCEPTABLE SUS SCORE? (Section 5.1, p. 591-592)

We believe that the rule-of-thumb standard discussed earlier, the so-called university grade analog, is well supported by both the adjective rating scale data and our own collective experience that takes into account the nature of the failures. This means that products which are at least passable have SUS scores above 70, with better products scoring in the high 70s to upper 80s. Truly superior products score better than 90. Products with scores of less than 70 should be considered candidates for increased scrutiny and continued improvement and should be judged to be marginal at best.

We can make a further distinction in the marginal scores, by dividing them into "low marginal" and "high marginal." This

break occurs roughly at the beginning of the second quartile range. Products with scores less than 50 should be cause for significant concern and are judged to be unacceptable.

Figure 13: Summary of Thresholds:

- NOT ACCEPTABLE: 0 to 50
- LOW MARGINAL: 50 to ~62.6
- HIGH MARGINAL: ~62.6 to 70
- ACCEPTABLE: 70 to 100